# LAMMPS official dem02-hertz-peak

Hertz compression and force at the source check time | 29 October 2026

| | |
|---|---|
| PASS | Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted. |

## Official quantities, errors and acceptance limits

| Quantity | Reference / limit | CPU dt | GPU dt | CPU dt/2 | GPU dt/2 |
|---|---|---|---|---|---|
| Overlap at official check time (m) | 0.0002741127<br>relative limit 0.01 | 0.0002741128<br>err 2.65e-07<br>PASS | 0.0002741128<br>err 2.65e-07<br>PASS | 0.0002741127<br>err 1.8e-07 PASS | 0.0002741127<br>err 1.8e-07 PASS |
| Force at official check time (N) | 10696.89<br>relative limit 0.01 | 10696.88<br>err 7.706e-07<br>PASS | 10696.88<br>err 7.706e-07<br>PASS | 10696.88<br>err 8.981e-07<br>PASS | 10696.88<br>err 8.981e-07<br>PASS |

Check time=3.016e-05 s; original step 1508. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

## Original bodies and source-to-native mapping

| Body | Radius / density | Initial xyz (m) | Initial v (m/s) | Initial spin (rad/s) |
|---|---|---|---|---|
| left | R=0.01<br>rho=2800 | (-0.0101, 0, 0) | (10, 0, 0) | (0, 0, 0) |
| right | R=0.01<br>rho=2800 | (0.0101, 0, 0) | (-10, 0, 0) | (0, 0, 0) |

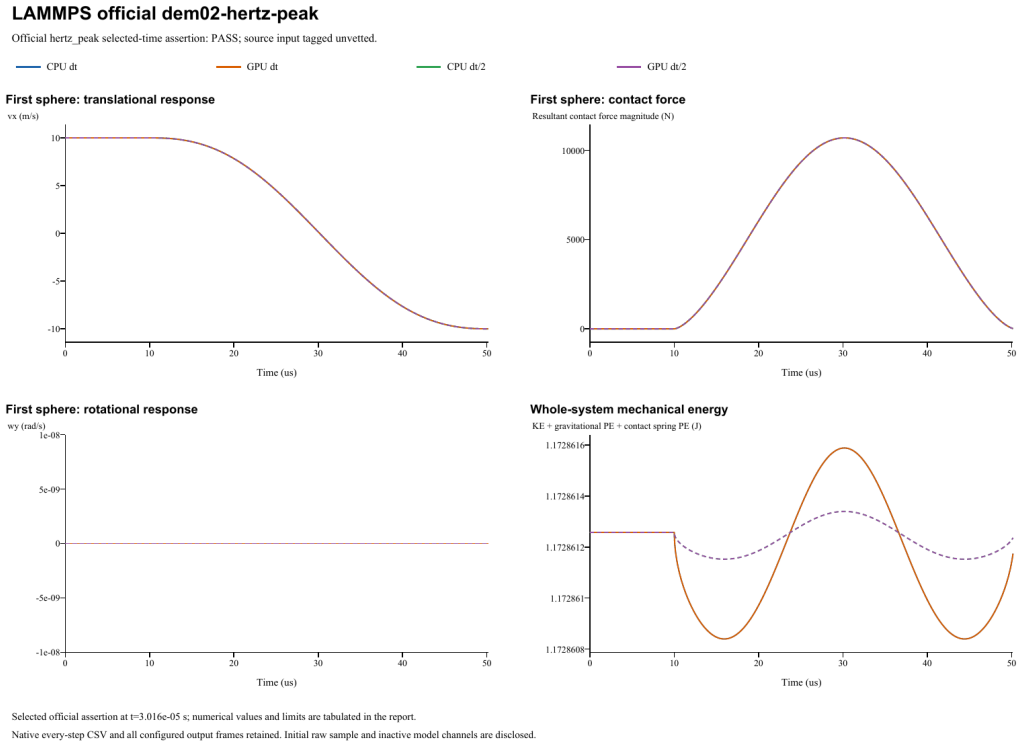
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=2.4999975e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=2e-08/1e-08 s; 2508/5016 steps. Direct Hertz kn mapped algebraically using E*=3kn/4 and equal-material E=2(1-nu^2)E*. Only normal channel active.

## Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 276bff16ac211f00347db6b5e20d23a25210e4ed. Official tag: unvetted.
LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

# LAMMPS native histories and refinement




## CPU/GPU and half-step comparison

| Quantity | CPU dt error | CPU dt/2 error | Error type / limit |
|---|---|---|---|
| Overlap at official check time (m) | 2.6498e-07 | 1.7997e-07 | relative: 0.01 |
| Force at official check time (N) | 7.7056e-07 | 8.9808e-07 | relative: 0.01 |

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 1.78e-15 m/s / 0 rad/s / 0 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 1.61e-15 kg m/s / 8.67e-19 m / 1.74e-15 m/s. Gravity is added separately to saved native contact forces.

## Complete results and verified scope

All 15,052 native frames are retained; main frame counts=[2509, 2509, 5017, 5017]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.